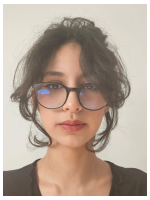
Mounji Malak



Engineering Student, Data Science · Big Data · Artificial Intelligence

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PORTFOLIO mounjimallak-bit.github.io/Portfolio: projects, code and live demos

PROFILE

Applying for a Master's degree in Data Science, Machine Learning or Artificial Intelligence, starting Fall 2027. I build ML systems that run on real data, from anomaly detection on sensor streams to monitoring and decision-support tools.

EDUCATION

ENSA Agadir, National School of Applied Sciences *Agadir, Morocco*
State Engineering Degree (Diplôme d'Ingénieur d'État), Data Science, Big Data & Artificial Intelligence *2022 to 2027 (expected)*
5 years at Master's level: 2 years of integrated preparatory classes, then 3 years of engineering cycle.
Relevant coursework: AI in Industrial Environments · Data Mining · AI Modeling · Sparsity in Machine Learning · AI Algorithms and Prediction · Big Data

IBN MAJA High School *Morocco*
Scientific Baccalaureate, With Highest Honors (18/20) *2022*
Mohammed VI Foundation Excellence Scholarship, awarded on Baccalaureate results, held 2022 to 2025 (full duration of the award).

EXPERIENCE

Machine Learning Engineer Intern (PFA) *July 2026 to August 2026*
Albarray Consulting, predictive maintenance for photovoltaic plants *Agadir, Morocco*
- Built an unsupervised anomaly detection pipeline for PV installations, trained on healthy operation data only and validated against labelled fault periods (shading) in the plant data provided for the internship.
- Combined Isolation Forest, a dense Autoencoder and an LSTM Autoencoder (ensembles of 5) by rank averaging: PR AUC 0.78 on the test set (Scikit-learn, TensorFlow/Keras).
- Designed the real time chain: Kafka ingestion, Python scoring consumer, TimescaleDB storage, dedicated alarm topic.
- Delivered an 8 panel Grafana dashboard provisioned as code; full stack containerised with Docker Compose.
  Code: github.com/mounjimallak-bit/pfa-photovoltaique

Data Analyst / Data Scientist Intern *July 2025*
HCP, High Commission for Planning *Morocco*
- Structured and harmonised public statistical tables into datasets ready for analysis and institutional decision making.
- Applied trend extrapolation models (Scikit-learn, linear regression) to socioeconomic indicator series.
- Built a decision support chatbot (Google Gemini API, prompt engineering, Streamlit) letting nontechnical users query indicators in natural language.
- Automated recurring analytical reporting (PDF: statistics, ranking tables, charts), replacing a manual spreadsheet process.

PROJECTS

Axiora Data, Socioeconomic Analytics Platform (Morocco) *2025*
Personal project extending my work at HCP: axiora-data.streamlit.app
- Streamlit application covering six thematic domains (population, labour, housing, finance, education, poverty), with choropleth maps at regional and provincial level (GeoPandas, Plotly).
- Automated PDF reporting (statistics, rankings, charts) and a natural language query layer over the indicators; deployed publicly.

Carrefour Sales Analysis · Python/Pandas ETL feeding a Power BI star schema (FactSales and 6 dimensions), with DAX measures and an interactive dashboard. Team project, 2025.

Real Estate Rental Platform · Flask and TailwindCSS web application with search, tenant reservations, owner dashboard, payments and reviews. Business logic in Oracle PL/SQL: stored procedures, functions and triggers for role-based access and non-overlapping reservations. 2025. Video demo available.

TECHNICAL SKILLS

Machine Learning	Scikit-learn · TensorFlow / Keras · Anomaly Detection (Isolation Forest, Autoencoder, LSTM Autoencoder) · Classification · Regression · Clustering · Random Forest · KNN · K-means
Deployment	Docker and Docker Compose · PostgreSQL / TimescaleDB · Grafana · Git / GitHub
Big Data & Streaming	Apache Kafka · Spark (SQL, Streaming, MLlib) · Hadoop · HDFS · Hive
Data & BI	Pandas · NumPy · ETL · Power BI (DAX, star schema) · Plotly · Matplotlib · Seaborn · GeoPandas · Streamlit
Programming	Python · SQL · PL/SQL · R · C
Databases & Tools	MySQL · Oracle SQL · MongoDB · Flask · Jupyter · LaTeX

LANGUAGES & ACTIVITIES

Languages	Arabic (Native) · French (C1) · English (C1)
Tutoring	Tutored 1st to 3rd year students at ENSA Agadir over several years
Volunteering	Amal Elkhayr association: renovation of a rural primary school near Agadir